

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEC602 - ELECTRICAL MACHINE DESIGN

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 A magnetic core has a length of 0.5 m and cross-sectional area of 5 cm². If a current of 3 A passes through a coil of 500 turns wound on it, and the relative permeability is 2000, what is the magnetic flux? (Assume $\mu_0 = 4\pi \times 10^{-7}$ H/m)
- a) 1.88×10^{-4} Wb b) 2.1×10^{-5} Wb c) 6.28×10^{-3} Wb d) 3.14×10^{-4} Wb

Computer-Aided Design (CAD) offers which of the following major advantages in electrical machine design?

- a) Reduces the number of poles b) Automates magnetic hysteresis measurement c) Enables iterative electromagnetic simulation d) Increases insulation thickness uniformly

- 3 A heating coil is to be designed for 1.5 kW at 230 V. What is the required resistance of the coil?
- a) 17.62 Ω b) 28.23 Ω c) 35.26 Ω d) 44.20 Ω

If a simplex lap winding has 36 conductors and 4 poles, the number of parallel paths is:

- a) 2 b) 4 c) 6 d) 12

A 3-phase induction motor has 40 kW output, 4 poles, 1440 rpm, and a specific electric loading of 25,000 A/m. Diameter is 0.25 m. Find the total ampere conductors per pole.

- a) 3925 b) 4925 c) 6125 d) 7850

- 6 An induction motor has a stator bore diameter of 0.25 m and a core length of 0.15 m. If specific magnetic loading $B_{av}=0.5$ T, specific electric loading $a_c=30,000$ A/m, and speed = 1440 rpm, find the output in kW using the output equation. Use $k_w=1$.

- a) 5.25 kW b) 6.6 kW c) 4.2 kW d) 7.5 kW

- 7 If the rotor bar current is 22 A, resistance per bar is 0.1Ω , and there are 40 bars in the cage, estimate total rotor copper loss.
a) 193.6 W b) 484.0 W c) 968.0 W d) 672.5 W
- 8 For an Induction Motor, rotor bar resistance of 0.04Ω and reactance of 0.3Ω , the rotor power factor at start is:
a) 0.12 b) 0.5 c) 0.13 d) 0.989
- 9 Which of the following is the main cause of heat generation in electrical machines?
a) Magnetic field distortion b) Electrical losses (I²R) c) Air friction d) Voltage regulation
- 10 If a machine has a heating time constant of 50 minutes, in how much time will it reach 90% of its final temperature?
a) 50 min b) 75 min c) 100 min d) 115 min
- 11 What happens to the no-load current if the core material has higher hysteresis loss?
a) Increases b) Decreases c) Remains constant d) Oscillates
- 12 The mechanical force during a short circuit in transformer windings is proportional to:
a) Voltage squared b) Current squared c) Core area d) Winding resistance

Q2. Solve the following.

1. Illustrate in brief principles of electrical machine design with a neat schematic. 6
2. Explain in brief the properties of electric and dielectric materials. 6

Q3. Solve the following.

- A) Draw the winding table for radial form for a 4 pole, 13 slot, simplex wave connected DC generator with a commutator having 13 segments. The number of coil sides per slot is 2 6
- B) Grade the resistance of a 5 section starter for a 7.5 kW, 250 V, 750 r.p.m. d.c. shunt motor from the following data: Maximum torque during starting period = 1.5 times full load torque; full load efficiency = 85.5 per cent. Armature circuit copper loss is 50 per cent of total loss. Field current = 2.6A, Find the speed at each stud when the notching takes place. 6

Q4. Solve Any Two of the following.

- A) Find the main dimensions, number of stator turns, size of conductor and number of stator slots of a 5 h.p., 400 V, 3-phase 4-pole squirrel cage induction motor using star-delta starter. Assume the following data: 6
1. Average flux density in the air gap = 0.46 Wb/m^2
 2. Ampere conductors per meter of armature periphery = 22×10^3
 3. Full load efficiency - 83 %
 4. Full load p.f. = 0.84 lagging
 5. Winding factor = 0.955
 6. Stacking factor = 0.9
 7. Current density = 4 A/mm^2
 8. No. of slots per poles per phase = 3
 9. $L/\tau = 1.5$

- B) A 15 HP, 400 volt, 1430 rpm, 3 phase induction motor with an efficiency of 80% and p.f. 81% has inner diameter of stator 30 cm and length 12 cm. Estimate the diameter and length for a 50 h.p., 406 V, 4 pole, 50 Hz induction motor to be designed for 84% efficiency and 85% p.f. 6
- C) Define specific magnetic loadings and specific electric loadings. Derive the output equation of an induction motor. 6

Q5. Solve Any Two of the following.

- A) A 11kW, 3 phase, 6 pole, 50 Hz, 220V star connected induction motor has 54 stator slots, each connecting 9 conductors. Calculate the values of bar and end ring currents. The number of rotor bar is 64. The machine has an efficiency of 0.86 and a power factor of 0.85. The rotor mmf may be assumed as 85 percent of stator mmf. Also find the bar and end ring sections if the current density is 5 A/mm². 6
- B) Elucidate in detail the factors affecting the selection of the length of the air gap of an induction motor. 6
- C) The temperature rise of a transformer is 25⁰ C after one hour and 37.5⁰ C after two hours, starting from cold conditions. Calculate its final steady temperature rise and heating time constant. 6

Q6. Solve Any Two of the following.

- A) Design magnetic frame of a 800kVA, 6.6kV/440V, three phase, 50Hz, core type, Delta-Star oil cooled transformer with respect to the following: 6
- Use the following data:
- Three stepped core ($K_i=0.6$)
 - $K=0.6$
 - Specific magnetic loading $B_m=1.5$ Wb/m²
 - Average current density, $\delta = 2.6$ A/mm²
 - The ratio of $\frac{L}{D-d} = 2.8$
 - Max. limit of temperature rise = 50⁰ c
 - Window space factor = $12/(30+kV)$
 - Cross sectional area of yoke, A_y 15% more than net iron area A_i
 - Specific resistance ρ , at 75⁰ c = 0.02 Ω /mm²/m
 - Use transformer steel lamination with a density of 7.55x10³ kg/m³ and iron losses of 1.6 watts/kg.
 - Diameter of cooling tube=5cm.
- Calculate kVA output of a single phase transformer from the following data: 6
- $\frac{\text{core height}}{\text{distance between core centres}} = 1.4$
 - $\frac{\text{diameter of circumscribing circle}}{\text{distance between core centres}} = 1.56$
 - $\frac{\text{net iron area}}{\text{area of circumscribing circle}} = 0.7$
 - current density = 2.3 $\frac{A}{mm^2}$
 - Window space factor = 0.27
 - frequency = 50 Hz
 - Flux density of core = 1.2 $\frac{wb}{m^2}$
 - Distance between core centres = 0.4m
- C) Explain in detail the mechanical forces that are generated in a transformer during short-circuit conditions. 6

*** End ***